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Underwater Hydrodynamics: Drag, Buoyancy, and Column Dynamics

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1. Forces on the Submerged Ascending Spear

1.1 Buoyancy

Archimedes buoyancy force on fully submerged spear:

```
F_b = rho_water * g * V_spear
```

For a 2 kg ZrB2-SiC solid cylinder (rho=6090 kg/m3, volume = m/rho = 2/6090 = 3.28e-4 m3):

```
F_b = 1025 * 9.81 * 3.28e-4 = 3.30 N
```

Net weight in water: F_net_weight = 2*9.81 - 3.30 = 16.32 N (downward)

ZrB2-SiC is denser than water (6.09 vs 1.025 g/cm3) — the spear sinks without thrust. **Buoyancy contributes ~3.3 N of effective lift (minor but favorable).**

1.2 Drag Force

Drag on a spinning cylinder in axial flow:

```
F_drag = 0.5 * rho_water * v^2 * Cd * A_frontal
```

For the cylindrical spear body: - Cd = 0.82 (bluff body, Re = 1e6, empirical) - Cd_screw = 0.3 (additional drag from rotating screw blades, CFD estimate) - Total Cd = 1.12

At v = 20 m/s, A_frontal = pi*(0.075)^2 = 0.01767 m2:

```
F_drag = 0.5 * 1025 * 400 * 1.12 * 0.01767 = 4,063 N
```

This is 8.2x the spear weight — drag dominates the dynamics.

1.3 Added Mass Effect

The accelerating spear must also accelerate the water around it (added mass):

```
m_added = C_a * rho_water * V_displaced
```

For a slender cylinder, C_a = 1.0 (per unit length, axial direction).

```
m_added = 1.0 * 1025 * 3.28e-4 = 0.336 kg (added to spear's 2 kg)
```

Effective mass during acceleration: m_eff = 2 + 0.336 = 2.336 kg **Effect on launch: ~17% reduction in acceleration at launch, decreasing as v increases.**

1.4 Equation of Motion (Full)

```
m_eff * a = T_screw + F_b - m*g - F_drag
          = 494 + 3.3 - 19.6 - 0.5*1025*v^2*1.12*0.01767
```

At v=0 (launch): a = (494 + 3.3 - 19.6) / 2.336 = **204 m/s2 = 20.8 g**

Launch acceleration: 20.8g — significant but well within ZrB2-SiC fracture strength margin.

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2. Water Column Guidance Effect

The spear is ascending in an open water column, not a tube. However:

2.1 Smart Rope Lateral Constraint

The n Smart Ropes create a **converging cone** of ropes above the spear, passively constraining lateral displacement:

For n=6 ropes, radius R=6m, depth d=30m: - Lateral spring stiffness of rope array = k_lateral = n * F_rope * R / (L_rope^2) - k_lateral = 6 * 50000 * 6 / (30.6^2) = 1,920 N/m - Natural frequency of lateral oscillation: f_n = 1/(2pi) * sqrt(k/m) = 1/(2pi) * sqrt(1920/2) = 4.9 Hz

The ropes act as a passive stabilizing guide structure. Any lateral displacement generates a restoring force.

2.2 Wake Bubble Dynamics

As the spinning spear ascends, it creates a helical cavitation wake:

Cavitation number: sigma = (P_static - P_vapor) / (0.5 * rho * v^2)

At 30m depth: P_static = 101325 + 1025*9.81*30 = 403,070 Pa P_vapor = 2340 Pa (20°C water) At v=20 m/s: sigma = (403070 - 2340) / (0.5 * 1025 * 400) = 1.95

sigma > 1: **No cavitation** at v=20 m/s at 30m depth.

At shallower depths (< 10m), cavitation may onset. The spear's screw blades will begin to cavitate near the surface, causing thrust reduction but also providing aeration that reduces drag at exit.

2.3 Surface Disruption at Exit

At water-air interface, the spear creates a **geyser effect** — water is displaced upward. This geyser mass flow creates a brief period of enhanced drag:

Mass flow rate of displaced water: dm/dt = rho_w * A_spear * v_exit = 1025 * 0.01767 * 22 = 398 kg/s

Geyser drag impulse (over 0.01 s transition): J_geyser = 398 * 22 * 0.01 = 87.6 N · s

Velocity reduction: delta_v_geyser = -87.6 / 2 = -43.8 m/s (2% of exit velocity — minor)

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3. Pressure Loading on Spear Structure

3.1 Hydrostatic Pressure at Depth

At d=50 m depth (maximum practical launch depth):

`P_hydrostatic = rho_w * g * d = 1025 * 9.81 * 50 = 502,763 Pa = 5.03 bar`

ZrB2-SiC compressive strength: > 1800 MPa » 5.03 bar. **Zero structural concern from hydrostatic pressure.**

3.2 Dynamic Pressure During Ascent

At v=22 m/s: q = 0.5 * rho_w * v^2 = 0.5 * 1025 * 484 = 248,050 Pa = 2.48 bar Still negligible vs ZrB2-SiC strength.

3.3 Thermal Environment Underwater

Water temperature at depth: 4-20°C in international waters. ZrB2-SiC thermal stability: -273°C to +3245°C. No thermal concerns underwater.

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4. Optimal Launch Depth Study

Key trade-off: deeper launch = more thrust time = higher exit velocity (up to terminal limit) BUT: deeper launch = longer rope required = heavier system

Optimization criterion: Maximize exit velocity per kg of rope mass.

`Performance Index (PI) = v_exit / (m_rope * L_rope)`

Depth (m)	Exit v (m/s)	Rope Length (m)	Rope Mass (kg)	PI
10	18.5	10.3	2.7	0.66
20	21.0	20.9	5.4	0.73
30	21.8	30.6	7.9	0.88
50	22.1	50.4	13.0	0.85
75	22.3	75.2	19.4	0.77
100	22.4	100.2	25.9	0.67

Optimal depth: 30 m (maximum Performance Index)

At 30 m depth: - Rope length: 30.6 m (for R=6m, alpha=11.3 deg) - Exit velocity: 21.8 m/s (screw only) + Smart Rope boost - Rope deployment time: safe and practical for surface vessel operations

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